

Group-5

Explainable Multi-Label Content-Based Image Retrieval for Aerial Images using Classical Computer Vision

Week-1: Work Done

Team Members

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GitHub Repository: [Link](#)

Week-1 Work

This week, the first stage of the development of our project was performed. It included the acquisition of the dataset needed for the development of our proposed system and the analysis of the research paper relevant to our problem statement. The main purpose of our project is to design an explainable Content-Based Image Retrieval (CBIR) system based on a classical computer vision approach, which retrieves images with visual and semantic similarities using handcrafted features.

The first key step of our project involved downloading the aerial image dataset for the development of our CBIR system. The images from this dataset include different types of objects and land-use classes. As aerial images can contain several semantic classes within a single image, our project will deal with multi-label image retrieval instead of classifying images into a single class. This dataset will serve as the foundation for further feature extraction, similarity evaluation, and retrieval assessment in our project.

The second key step this week involved analysing the research paper related to our project. The paper was studied to understand the main principles of Content-Based Image Retrieval.

Particular attention was given to the contribution of handcrafted visual features that encode various image properties, such as colour, texture, and spatial properties. Such visual features are essential because they allow images to be represented numerically, making it possible to compare images in terms of their visual properties without applying deep learning methods.

In addition, this work helped us understand the significance of explainability within CBIR systems. Rather than considering image retrieval as a black-box process, our

proposed system aims to provide an interpretable explanation for the retrieval of images. This can be achieved by analysing the contribution of each handcrafted feature to the similarity between a query image and the retrieved images.